

Assefa Wahd

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Education

University of Alberta <i>PhD in Radiology and Diagnostic Imaging</i>	<i>Sept 2023 – present</i>
◦ GPA: 3.7/4.0	
University of Science and Technology (UST), South Korea <i>MEng in Computer Software</i>	<i>Sept 2021 – Aug 2023</i>
◦ GPA: 4.15/4.5	
University of Elec. Science and Tech. of China (UESTC), China <i>BEng in Electronic Information Engineering</i>	<i>Sept 2017 – July 2021</i>
◦ GPA: 4.0/4.0	

Research interest

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- Computer vision
 - Reinforcement learning
 - Medical imaging

Honors and Awards

Alberta Bench to Bedside Young Innovators Award – 2nd prize <i>Bench to Bedside Students' Association, University of Alberta</i>	<i>Mar 23 2024</i>
Faculty of Medicine & Dentistry 75th Anniversary Award <i>University of Alberta</i>	<i>Jan 2024 - Dec 2024</i>
Faculty of Medicine & Dentistry Delnor Scholarship <i>University of Alberta</i>	<i>Jan 2024 - Dec 2024</i>
Graduation Research Assistantship <i>University of Alberta</i>	<i>Sep 2023 - Jun 2027</i>
Graduate Student Scholarship <i>Korea National University of Science and Technology (UST)</i>	<i>Sep 2021 - Aug 2023</i>
Academic Achievement Award <i>University of Science and Technology of China (UESTC)</i>	<i>2019 - 2020</i>
Academic Achievement Award <i>University of Science and Technology of China (UESTC)</i>	<i>2018 - 2019</i>
Academic Achievement Award <i>University of Science and Technology of China (UESTC)</i>	<i>2017 - 2018</i>
Outstanding Academic Achievement Award <i>Ethiopian Chinese Chamber of Commerce</i>	<i>2016</i>

Experience

Graduate Research Assistant <i>University of Alberta</i>	<i>Edmonton, Alberta</i> <i>Sep 2023 – present</i>
Deep Learning and Reinforcement Learning (DLRL) Summer School <i>CIFAR</i>	<i>Edmonton, Alberta</i> <i>Jul 21 – Aug 01</i>
◦ Attended the 2025 CIFAR Deep Learning + Reinforcement Learning Summer School	
◦ Worked on maximum entropy reinforcement learning project	

AI Accelerator Program Participant

Alberta Machine Intelligence Institute (Amii)

- Contributed 20 hours+ of *Work-Integrated Learning Opportunity (WILO)*
- Built a case study on *semantic segmentation* with a short-term, part-time contract

Edmonton, Alberta

Oct 2024 – Nov 2024

Student Researcher

Electronics and Telecommunications Research Institute (ETRI)

Daejeon, South Korea

Sep 2021 – Aug 2023

Project Title: *Development of Grasp Manipulation Technology for Flexible Cable Wiring.*

- Implemented several bottom-up approaches for instance segmentation of overlapping industrial wires (metric learning-, affinity-, reinforcement learning-, and RNN-based techniques).
- Developed a new data augmentation technique that exponentially increases the number of ground truth segmentations.
- Experience working with limited data; Semi-Supervised Learning (SSL), Weakly Supervised Object Localization (WSOL), Semantic (WSSS) and Instance Segmentation (WSIS).

Project Title: *Open-World Object Detection.*

- The goal of the project is to predict bounding boxes both known and unknown objects in images. Most object detection models treat objects with no groundtruth as background, however, in real-world we need to know both known and unknown objects (open world).
- I implemented several region proposal networks for open-world object detection ([OLN RPN](#) [↗](#), [MVIT RPN](#) [↗](#), [SAM RPN](#) [↗](#)).
- (*Ongoing project*) Unknown object detection using YOLO v5/v8 with [SAM](#) [↗](#) proposals.
- (*Ongoing project*) Unknown object detection using Detection Transformer (DETR) with [SAM](#) [↗](#) Proposals.

Project Title: *Instance Segmentation of Cables using Actor-Critic Methods.*

- Proposed a novel way that models instance segmentation as a multimodal image-to-image translation (I2I) and a reinforcement learning problem in a continuous action space.
- Developed a working instance segmentation model using [BicycleGAN](#) [↗](#), and actor-critic methods.

Master's Thesis: *On the effectiveness of synthetic data for out-of-distribution detection.*

- Use generative models (StyleGAN2, Diffusion Models, Energy-Based Models) to generate synthetic OOD data
- Evaluate several classification models for out-of-distribution detection
- The models considered include WideResNet, DenseNet, MLP-mixer, ViT-B/16, ViT-L/16.

Publications

[Time-Contrastive Pretraining for In-Context Image and Video Segmentation](#) [↗](#)

Feb 2025

Assefa Wahd, Jacob L. Jaremko , Abhilash Hareendranathan

International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Springer Nature Switzerland, 2025.

[Sam2Rad: A Segmentation Model for Medical Images with Learnable Prompts](#) [↗](#)

Aug 2024

Assefa Wahd, Banafshe Felfeliyan, Yuyue Zhou, Shrimanti Ghosh, Adam McArthur, Jiechen Zhang, Jacob L. Jaremko, Abhilash Hareendranathan
Computers in Biology and Medicine 187 (2025).

Shoulder Rotator Cuff Tear Detection from Ultrasound Videos using Deep Reinforcement Learning

Sep 2024

Shrimanti Ghosh, Geetika Vadali, Ayush Singh, Banafshe Felfeliyan, Yuyue Zhou, **Assefa Wahd**, Jessica Knight, Mahesh R. Panicker, Jacob L. Jaremko, Jacob L. Jaremko, Abhilash Hareendranathan

Submitted to “(ISBI 2025) 2025 IEEE International Symposium on Biomedical Imaging”.

Deep metric learning-based Out-of-Distribution Detection with Synthetic Outlier Exposure [🔗](#) Aug 2024

Semantic AutoSAM: Self-Prompting Segment Anything Model for Semantic Segmentation of Medical Images Dec 2023

Assefa Wahd, Jessica Küpper, Jacob L. Jaremkó, Abhilash R. Hareendranathan
The IEEE Engineering in Medicine and Biology Society (EMBC 2024).

Cable Instance Segmentation with Synthetic Data Generation [🔗](#) Jan 2023

Assefa Wahd, Donghyung Kim, Seung-Ik Lee
The 21st International Conference on Control, Automation and Systems (ICCAS 2022) (Bexco, Pusan, Korea).

doi: 10.23919/ICCAS55662.2022.10003680 [🔗](#)

Low-Likelihood EBM Samples for Out-of-Distribution Detection Feb 2023

Assefa Wahd, Marcella Astrid, Seung-Ik Lee
Accepted to “The 35th Workshop on Image Processing and Image Understanding (IPIU 2023) (Jeju, Korea)”.

[GDrive](#) [🔗](#)

Joint Classification and Normalizing Flow Model for Out-of-Distribution Detection Jan 2023

Assefa Wahd, Donghyung Kim, Seung-Ik Lee
Accepted to “The electronics and information engineers (IEIE 2023) (Jeju, Korea)”.

[DBpia](#) [🔗](#)

Leveraging Class-agnostic Object Proposals for Out-of-Distribution Object Detection June 2023

Assefa Wahd, Minsu Jang, Seung-Ik Lee
Accepted to “Learning by Asking for Intelligent Robots and Agents”.

Skills

Programming Languages: Python, C, Bash, HTML, CSS

ML/DL: PyTorch, Scikit-learn, OpenCV

Languages: English (IELTS: 8.0), Chinese, Tigrinya, Amharic